1. **List the key differences between IPv4 and IPv6 addressing by creating a two-column table with at least 4 points for each.**

IPv4	**IPv6**
It Uses 32-bit addresses.	It uses 128-bit addresses.
IPv4 Address format is decimal.	IPv6 Address format is hexadecimal.
It supports about 4.3 billion unique IP addresses.	It supports almost unlimited unique IP addresses.
It uses NAT because of limited address space.	Generally it does not require NAT due to the vast address space.

2. **Given the following scenario: You have 4 devices (laptop, phone, smart TV, and printer) connected to your home Wi-Fi. Assign a unique static IPv4 address to each device using the 192.168.10.0/24 network, and write down the IP, subnet mask, and default gateway for each device.**

- Here, 192.168.10.0/24 is given, from that:
- Network Address is 192.168.10.0
- Subnet Mask is 255.255.255.0
- And default gateway is 192.168.10.1
- I have created a table that describes static IP addresses assigns to the four devices.

Device	Static IP Address	Subnet Mask	Default Gateway
Laptop	192.168.10.10	255.255.255.0	192.168.10.1
Phone	192.168.10.11	255.255.255.0	192.168.10.1
Smart TV	192.168.10.12	255.255.255.0	192.168.10.1
Printer	192.168.10.13	255.255.255.0	192.168.10.1

3. **Calculate the number of valid subnets and hosts per subnet if you use a /27 CIDR subnet mask on a 192.168.1.0 network. List the first three valid subnet address ranges (network address to broadcast address) and the usable host IPs in each.**

Step-1 Number of Valid subnets

- Original CIDR: /24
- New CIDR: /27
- 3 bits borrowed from the last octant = 27-24 = 3
- Number of valid subnets are 2^3 = 8 subnets.
- So, valid subnets are now 8.

Step-2 Calculation Hosts per subnet

- There are total 32 bits in the IP address. From the 32 bit IP address I subtract new CIDR = 32-27 = 5
- Total addresses per subnet is 2^5= 32
- So, usable host addresses are now 32-2 = 30 hosts.

First Three Valid Subnets

Subnet	Network Address	First Host	Last Host	Broadcast Address
1	192.168.1.0/27	192.168.1.1	192.168.1.30	192.168.1.31
2	192.168.1.32/27	192.168.1.33	192.168.1.62	192.168.1.63
3	192.168.1.64/27	192.168.1.65	192.168.1.94	192.168.1.95

First Subnet Range

- Network address – 192.168.1.0
- Usable hosts – 192.168.1.1 – 192.168.1.30
- Broadcast address – 192.168.1.31

Second Subnet Range

- Network address – 192.168.1.32

- Usable hosts – 192.168.1.33 – 192.168.1.62
- Broadcast address – 192.168.1.63

Third Subnet Range

- Network address – 192.168.1.64
- Usable hosts – 192.168.1.65 – 192.168.1.94
- Broadcast address – 192.168.1.95

4. **You are given the IP address 10.0.5.17 with subnet mask 255.255.255.0. Determine the network address, broadcast address, and the range of valid host addresses for this subnet.**

- Here, IP address is 10.0.5.17
- And subnet mask is 255.255.255.0

Step 1 conversion of subnet mask into the CIDR

- 255.255.255.0 = /24
- /24 which means, first 24 bits are network bits and 8 bits are for hosts.

Step 2 Calculation of Network Address

- IP address is – 10.0.5.17
- Subnet mask is – 255.255.255.0
- So, Network address will be 10.0.5.0

Step 3 Calculation of Broadcast Address

- For a /24 CIDR network, all host bits are set to 1:
- Which means Broadcast address will be 10.0.5.255

Step 4 Valid Host Range

- First usable host is 10.0.5.1
- And last usable host is 10.0.5.254

5. **Use an online subnet calculator tool (such as https://www.subnet-calculator.com/) to verify your answer for the previous task. Take a screenshot of your calculation and write one line explaining if your manual answer matched the tool's output.**

- Here, IP address is 10.0.5.17
- And subnet mask is 255.255.255.0

Step 1 conversion of subnet mask into the CIDR

- 255.255.255.0 = /24
- /24 which means, first 24 bits are network bits and 8 bits are for hosts.

Step 2 Calculation of Network Address

- IP address is – 10.0.5.17
- Subnet mask is – 255.255.255.0
- So, Network address will be 10.0.5.0

Step 3 Calculation of Broadcast Address

- For a /24 CIDR network, all host bits are set to 1:
- Which means Broadcast address will be 10.0.5.255

Step 4 Valid Host Range

- First usable host is 10.0.5.1
- And last usable host is 10.0.5.254

- I verified the subnet using an online subnet calculator. The tool’s output matched my manual calculation which was, Network address = 10.0.5.0, Broadcast address = 10.0.5.255 and Valid Host Range = 10.0.5.1 to 10.0.5.254